

# AIXI

David Quarel

Australian National University

July 16, 2026

## What you'll learn

- Formalize the agent-environment interaction, value functions, and the Bayes-optimal policy that defines AIXI.
- Derive the explicit expectimax form of AIXI.
- Prove on-policy value convergence and that AIXI cannot be fooled in deterministic environments.
- Prove the self-optimizing property via likelihood-ratio martingales and change of measure.

Much of this material is drawn from [?]. The book provides fuller explanations, additional context, and proofs omitted here.

**Difficulty ratings.** Each subproblem is tagged with a rating in square brackets using Knuth's scale [?]: roughly, [00] trivial, [10] 15-minute pencil-and-paper, [20] 1–2 hours, [30] several hours to a day, [40] a significant research result. Intermediate values are possible. See Appendix C for the full scale.

Problems marked (\*) are less interesting/insightful and while the result may be used later, I would recommend skipping them on a first pass.

## Overview

AIXI is the *action* half of **Universal AI**: it lifts the Bayesian mixture from [Solomonoff induction](#) — learning to predict — to learning to act. Now the agent takes actions that shape what it observes, and AIXI is the **Bayes-optimal policy** over a universal mixture  $\xi$  of environments. It learns to act as well as if it knew the true environment  $\mu$ . Under the universal choice —  $\mathcal{M}$  = all lower-semicomputable environments, prior  $w_\nu = 2^{-K(\nu)}$  — the assumption “ $\mu \in \mathcal{M}$ ” again becomes “the universe is computable.”

## Prerequisites

- [Solomonoff Induction](#) — **start there first.** The Bayesian mixture, posterior updates, and dominance carry over directly; AIXI reuses them in the action setting.
- Comfort with discrete probability (conditional distributions, chain rule, expectations).
- Familiarity with sequential decision-making / reinforcement learning (agents, rewards, discounting, value functions) is useful but not assumed.

## Further reading

- Hutter, [An Introduction to Universal Artificial Intelligence](#) (2024): Chapter 2.7 (Kolmogorov complexity), Chapters 3.7–3.8 (the model class and universal prior), and Chapter 7.4 (AIXI).
- Hutter, [Universal Artificial Intelligence: Sequential Decisions Based on Algorithmic Probability](#) (Springer, 2005) — the original book-length treatment; Lem. 5.28 handles the countable- $\mathcal{M}$  self-optimizing case.
- Blackwell & Dubins, [Merging of Opinions with Increasing Information](#) (Ann. Math. Statist., 1962) — the merging-of-opinions theorem behind on-policy value convergence.
- Leike & Hutter, [Bad Universal Priors and Notions of Optimality](#) (COLT 2015) — adversarial choices of the universal Turing machine can make AIXI behave arbitrarily badly.

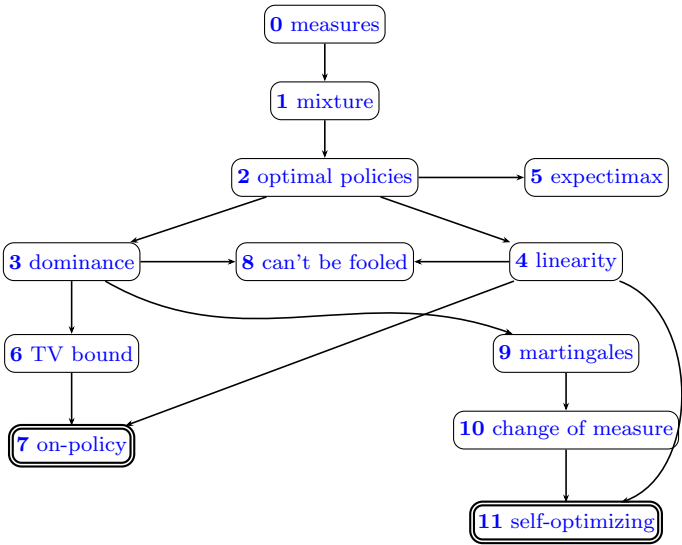
## Goal and Roadmap

This exercise sheet builds towards three main results:

- **On-policy value convergence** (Section 7): the Bayesian mixture  $\xi$  learns to predict the value of any fixed policy as well as the true environment  $\mu$ .
- **AIXI can't be fooled** (Section 8): in deterministic environments, the Bayes-optimal agent is guaranteed non-zero value whenever optimal value is non-zero.
- **Self-optimizing property** (Section 11, advanced stretch goal): the Bayes-optimal policy  $\pi_\xi^*$  learns to *act* as well as if it knew  $\mu$ , provided that any learnable policy can achieve this. This is the central theoretical justification for the AIXI agent.

A good target is to complete Sections 7 and 8. The self-optimizing property (Sections 9–11) requires substantial additional machinery (supermartingales, change of measure) and is an advanced stretch goal.

**Critical path.** The three results share a common foundation (Sections 0–4) and then diverge:



**In summary:**

- Sections 0–4 establish the Bayesian RL framework: measure algebra, mixture properties, existence of optimal policies, dominance, and linearity.
- Section 5 derives the explicit expectimax form of AIXI.
- Sections 6–7 prove *on-policy value convergence*:  $V_\xi^\pi$  and  $V_\mu^\pi$  become indistinguishable for any fixed  $\pi$ .
- Section 8 shows *AIXI can't be fooled*: the Bayes-optimal agent achieves non-zero value whenever optimal value is non-zero (in deterministic environments).
- Sections 9–11 (advanced stretch goal) prove the *self-optimizing property*: if any policy can learn to act optimally,  $\pi_\xi^*$  inherits this.

## Setup

An **agent** interacts with an **environment** in discrete time steps  $t = 1, 2, \dots$

**Definition 0.1** (Spaces and notation).

- $\mathcal{A}$ : finite set of **actions**
- $\mathcal{O}$ : finite set of **observations**

- $\mathcal{R} \subset [0, 1]$ : finite set of **rewards**
- $\mathcal{E} := \mathcal{O} \times \mathcal{R}$ : set of **percepts**;  $e_t = (o_t, r_t) \equiv o_t r_t$
- $\mathcal{H}^t := (\mathcal{A} \times \mathcal{E})^t$ : set of all histories of length  $t$
- $\mathcal{H}^* := \cup_{t=0}^{\infty} \mathcal{H}^t$ : set of all finite **histories**
- $\mathcal{H}^\infty := (\mathcal{A} \times \mathcal{E})^\infty$ : set of all infinite histories
- $\Delta S$ : set of all probability distributions over set  $S$
- $\llbracket \cdot \rrbracket$ : **Iverson bracket**:  $\llbracket P \rrbracket = 1$  if  $P$  is true, 0 if false
- $t$ : current time step;  $m$ : finite horizon;  $1 \leq t \leq m$
- $i, j, k$ : arbitrary integer indices
- $xy$ : concatenation
- $\epsilon$ : the empty string/history
- $\mathfrak{x}_{i:j} := a_i e_i a_{i+1} e_{i+1} \cdots a_j e_j$ : history segment from time  $i$  to  $j$
- $\mathfrak{x}_{<t} := a_1 e_1 a_2 e_2 \cdots a_{t-1} e_{t-1}$ : history up to (but not including) time  $t$

**Definition 0.2** (Policy  $\pi$ ). A **policy**  $\pi : \mathcal{H} \rightarrow \Delta \mathcal{A}$  maps each history to a probability distribution over actions. Given history  $\mathfrak{x}_{<t}$ :

- $\pi(\cdot \mid \mathfrak{x}_{<t})$  is a distribution over  $\mathcal{A}$ ,
- $\pi(a_t \mid \mathfrak{x}_{<t}) \in [0, 1]$  is the probability of choosing action  $a_t$ ,
- the agent samples  $a_t \sim \pi(\cdot \mid \mathfrak{x}_{<t})$ .

A policy is **deterministic** if  $\pi(a \mid \mathfrak{x}_{<t}) \in \{0, 1\}$  for all  $a, \mathfrak{x}_{<t}$ . We write  $\pi(\mathfrak{x}_{i:j}) := \prod_{k=i}^j \pi(a_k \mid \mathfrak{x}_{<k})$ .

**Definition 0.3** (Environment  $\nu$ ). An **environment**  $\nu : \mathcal{H} \times \mathcal{A} \rightarrow \Delta \mathcal{E}$  maps each history–action pair to a distribution over percepts. Given history  $\mathfrak{x}_{<t}$  and action  $a_t$ :

- $\nu(\cdot \mid \mathfrak{x}_{<t} a_t)$  is a distribution over  $\mathcal{E}$ ,
- $\nu(e_t \mid \mathfrak{x}_{<t} a_t) \in [0, 1]$  is the probability of percept  $e_t$ ,
- the environment samples  $e_t \sim \nu(\cdot \mid \mathfrak{x}_{<t} a_t)$ .

We write  $\nu(\mathfrak{x}_{i:j}) := \prod_{k=i}^j \nu(e_k \mid \mathfrak{x}_{<k} a_k)$ . This satisfies the chain rule:  $\nu(\mathfrak{x}_{i:j}) = \nu(\mathfrak{x}_{i:j-1}) \cdot \nu(e_j \mid \mathfrak{x}_{i:j-1} a_j)$  or in the form we will usually use,  $\nu(\mathfrak{x}_{1:t}) = \nu(\mathfrak{x}_{<t}) \cdot \nu(e_t \mid \mathfrak{x}_{<t} a_t)$ . An environment is **deterministic** if  $\nu(e_t \mid \mathfrak{x}_{<t} a_t) \in \{0, 1\}$  for all  $e_t, \mathfrak{x}_{<t}, a_t$  (each percept is produced with certainty). We denote the true (unknown) environment by  $\mu$ .

**Definition 0.4** (Interaction measure  $\nu^\pi$ ). When policy  $\pi$  interacts with environment  $\nu$ , the joint probability of a history segment  $\mathfrak{a}_{i:j}$  given past  $\mathfrak{a}_{< i}$  is

$$\nu^\pi(\mathfrak{a}_{i:j} \mid \mathfrak{a}_{< i}) := \prod_{k=i}^j \pi(a_k \mid \mathfrak{a}_{< k}) \nu(e_k \mid \mathfrak{a}_{< k} a_k).$$

## 0. Properties of Measures

**Exercise 0.1 (Factorization).** [05] Show that  $\nu^\pi(\mathfrak{a}_{1:t}) = \pi(\mathfrak{a}_{1:t}) \cdot \nu(\mathfrak{a}_{1:t})$ .

**Exercise 0.2 (Chain rule).** [05] Show that  $\nu(\mathfrak{a}_{1:t}) = \nu(\mathfrak{a}_{< t}) \cdot \nu(e_t \mid \mathfrak{a}_{< t} a_t)$ .

**Exercise 0.3 (Marginalizing percepts).** [03] (\*) Show that  $\sum_{e_t} \nu^\pi(a_t e_t \mid \mathfrak{a}_{< t}) = \pi(a_t \mid \mathfrak{a}_{< t})$ .

**Exercise 0.4 (Deterministic interaction measure).** [05] (\*) Let  $\pi$  be a deterministic policy, i.e. at each history  $\mathfrak{a}_{< k}$  there is a unique action  $a_k^*$  with  $\pi(a_k^* \mid \mathfrak{a}_{< k}) = 1$ . Show that, provided  $\mathfrak{a}_{i:j}$  is consistent with  $\pi$  (i.e.  $a_k = a_k^*$  for every  $k \in \{i, \dots, j\}$ ):

$$\nu^\pi(\mathfrak{a}_{i:j} \mid \mathfrak{a}_{< i}) = \nu(\mathfrak{a}_{i:j} \mid \mathfrak{a}_{< i}),$$

i.e. on  $\pi$ -consistent futures the interaction measure reduces to the environment measure: every policy factor is 1.

**Exercise 0.5.** [05] (General chain rule for  $\nu^\pi$ ) Show that for any  $p \leq q \leq r$ :

$$\nu^\pi(\mathfrak{a}_{p:r} \mid \mathfrak{a}_{< p}) = \nu^\pi(\mathfrak{a}_{p:q} \mid \mathfrak{a}_{< p}) \cdot \nu^\pi(\mathfrak{a}_{q+1:r} \mid \mathfrak{a}_{< p} \mathfrak{a}_{p:q}).$$

## The Bayesian Mixture and Value Function

**Definition 0.1** (Model class, prior, and Bayesian mixture  $\xi$ ). Let  $\mathcal{M} = \{\nu_1, \nu_2, \dots\}$  be a countable class of environments with prior weights  $w_\nu > 0$  satisfying  $\sum_{\nu \in \mathcal{M}} w_\nu \leq 1$ . We assume  $\mu \in \mathcal{M}$ .

The **Bayesian mixture**  $\xi$  is defined as

$$\xi(\mathfrak{a}_{1:t}) := \sum_{\nu \in \mathcal{M}} w_\nu \nu(\mathfrak{a}_{1:t}) \quad \text{and} \quad \xi(e_t \mid \mathfrak{a}_{< t} a_t) := \xi(\mathfrak{a}_{1:t}) / \xi(\mathfrak{a}_{< t}).$$

One can show that the one-step predictive distribution can be written as

$$\xi(e_t \mid \mathfrak{a}_{<t} a_t) = \sum_{\nu \in \mathcal{M}} w(\nu \mid \mathfrak{a}_{<t}) \nu(e_t \mid \mathfrak{a}_{<t} a_t),$$

where the **posterior weight** is

$$w(\nu \mid \mathfrak{a}_{<t}) := w_\nu \frac{\nu(\mathfrak{a}_{<t})}{\xi(\mathfrak{a}_{<t})}, \quad w(\nu \mid \epsilon) := w_\nu.$$

See [Appendix A](#) for a worked example.

**Definition 0.2** (Expectation  $\mathbb{E}_\nu^\pi$ ). For a function  $f : \mathcal{H} \rightarrow \mathbb{R}$  of a finite future segment  $\mathfrak{a}'_{t:m}$ :

$$\mathbb{E}_\nu^\pi[f \mid \mathfrak{a}_{<t}] := \mathbb{E}_{\mathfrak{a}_{t:m} \sim \nu^\pi(\cdot \mid \mathfrak{a}_{<t})}[f] = \sum_{\mathfrak{a}'_{t:m}} \nu^\pi(\mathfrak{a}'_{t:m} \mid \mathfrak{a}_{<t}) f(\mathfrak{a}'_{t:m}).$$

For functions  $f$  of the infinite future (like the value function), with a sequence  $f_1, f_2, \dots \rightarrow f$ , where  $f_m$  depends only on  $\mathfrak{a}'_{t:m}$ , we define  $\mathbb{E}_\nu^\pi[f \mid \mathfrak{a}_{<t}] := \lim_{m \rightarrow \infty} \mathbb{E}_\nu^\pi[f_m \mid \mathfrak{a}_{<t}]$ ,

**Definition 0.3** (Discounted return  $G_{t:m}$ ). Fix a discount factor  $\gamma \in (0, 1)$ , held constant throughout. The **discounted return** at time step  $t$ , up to horizon  $m$ , is

$$G_{t:m}^\gamma := \sum_{k=t+1}^m \gamma^{k-(t+1)} r_k = r_{t+1} + \gamma r_{t+2} + \dots + \gamma^{m-t} r_m.$$

Since  $\gamma$  is fixed we drop it and write  $G_{t:m} \equiv G_{t:m}^\gamma$ . It satisfies the recursion  $G_{t:m} = r_{t+1} + \gamma G_{t+1:m}$ , with  $G_{m-1:m} = r_m$  and  $G_{n:m} = 0$  for  $n \geq m$ . The **infinite-horizon return** is the pointwise limit

$$G_{\geq t} \equiv G_{t:\infty} := \lim_{m \rightarrow \infty} G_{t:m} = \sum_{k=t+1}^{\infty} \gamma^{k-(t+1)} r_k,$$

with the clean recursion  $G_{\geq t} = r_{t+1} + \gamma G_{\geq t+1}$  and no boundary cases.

**Definition 0.4** (Value function<sup>a</sup>). The **value** of policy  $\pi$  in environment  $\nu$  with horizon  $m \geq t$  given history  $\mathfrak{a}_{<t}$  is

$$\begin{aligned} V_\nu^{\pi,m}(\mathfrak{a}_{<t}) &:= (1 - \gamma) \mathbb{E}_\nu^\pi[G_{t-1:m} \mid \mathfrak{a}_{<t}] \\ &= (1 - \gamma) \sum_{\mathfrak{a}_{t:m}} \nu^\pi(\mathfrak{a}_{t:m} \mid \mathfrak{a}_{<t}) G_{t-1:m} \\ &= (1 - \gamma) \sum_{\mathfrak{a}_{t:m}} \nu^\pi(\mathfrak{a}_{t:m} \mid \mathfrak{a}_{<t}) \left[ \sum_{k=t}^m \gamma^{k-t} r_k \right]. \end{aligned}$$

The **infinite-horizon value** is the pointwise limit  $V_\nu^{\pi, \infty}(\mathfrak{a}_{<t}) := \lim_{m \rightarrow \infty} V_\nu^{\pi, m}(\mathfrak{a}_{<t}) = (1 - \gamma) \mathbb{E}_\nu^\pi[G_{\geq t-1} \mid \mathfrak{a}_{<t}]$ . We write  $V_\nu^\pi \equiv V_\nu^{\pi, \infty}$  for short. The **optimal value** is  $V_\nu^{*, m}(\mathfrak{a}_{<t}) := \sup_\pi V_\nu^{\pi, m}(\mathfrak{a}_{<t})$  for  $m \in \mathbb{N} \cup \{\infty\}$ , with  $V_\nu^* \equiv V_\nu^{*, \infty}$ .<sup>b</sup> An **optimal policy**  $\pi_\nu^*$  satisfies  $V_\nu^{\pi_\nu^*} = V_\nu^*$ . The **Bayes-optimal policy** is  $\pi_\xi^* \in \arg \max_\pi V_\xi^\pi$ .

<sup>a</sup>See [Appendix B](#) for a worked example. For simplicity, we consider only geometric discounting.

<sup>b</sup>The optimal value is defined as a sup over policies. In general, a supremum need not be attained (e.g.  $\sup_{x \in (0,1)} x = 1$  but no  $x \in (0,1)$  achieves it). Section 2 shows that the sup is attained in our setting.

## Note

**AIXI.** All results in this sheet hold for any countable  $\mathcal{M}$  and prior weights  $w_\nu$ . **AIXI** is a special case of the Bayes-optimal policy  $\pi_\xi^*$  for the particular choice  $\mathcal{M} :=$  the class of all lower-semicomputable chronological semimeasures, and prior  $w_\nu := 2^{-K(\nu)}$ , where  $K(\nu)$  is the Kolmogorov complexity of  $\nu$ : the length of the shortest program that computes  $\nu$ . The resulting agent is written  $\pi_{\xi_U}^*$ , or simply  $\text{AIXI}$ .

**Why this  $\mathcal{M}$ ?** By including every computable environment, the assumption  $\mu \in \mathcal{M}$  reduces to “the universe is computable”: as weak an assumption as one can make.

**Why this prior?** The prior  $2^{-K(\nu)}$  is *dominant*: for any other computable prior  $w'_\nu$ , there exists a constant  $c > 0$  such that  $2^{-K(\nu)} \geq c \cdot w'_\nu$  for all  $\nu$ . S

**Caveat:** The constant  $c$  depends on the choice of universal Turing machine  $U$ , and adversarial choices of  $U$  can make AIXI behave arbitrarily badly [?].

See [?]: Chapter 2.7 for Kolmogorov complexity, Chapters 3.7–3.8 for the model class and universal prior, and Chapter 7.4 for AIXI itself.

## 1. Properties of the Bayesian Mixture

**Exercise 1.1 (Posterior update).** [10] (\*) Show that the posterior updates multiplicatively:

$$w(\nu \mid \mathfrak{a}_{1:t}) = w(\nu \mid \mathfrak{a}_{<t}) \frac{\nu(e_t \mid \mathfrak{a}_{<t} a_t)}{\xi(e_t \mid \mathfrak{a}_{<t} a_t)}.$$

**Hint:** Use the definitions of  $w(\nu \mid \mathfrak{a}_{1:t})$  and  $w(\nu \mid \mathfrak{a}_{<t})$ , and apply Exercise 0.2.

**Exercise 1.2.** [15] (One-step predictive distribution of  $\xi$ ) Starting from  $\xi(\mathfrak{a}_{1:t}) = \sum_{\nu \in \mathcal{M}} w_\nu \nu(\mathfrak{a}_{1:t})$ , derive the one-step predictive form:

$$\xi(e_t \mid \mathfrak{x}_{<t} a_t) = \sum_{\nu \in \mathcal{M}} w(\nu \mid \mathfrak{x}_{<t}) \nu(e_t \mid \mathfrak{x}_{<t} a_t).$$

**Hint:** Write  $\xi(e_t \mid \mathfrak{x}_{<t} a_t) = \xi(\mathfrak{x}_{1:t}) / \xi(\mathfrak{x}_{<t})$ , expand the numerator, and use Exercise 0.2.

**Exercise 1.3 (Bounded value).** [10] Show that  $V_\nu^{\pi, m}(\mathfrak{x}_{<t}) \in [0, 1]$  for any  $\nu, \pi, m \in \mathbb{N} \cup \{\infty\}, \mathfrak{x}_{<t}$ .

**Hint:** Recall the formula for geometric series:  $(1 - \gamma) \sum_{k=0}^n \gamma^k = 1 - \gamma^{n+1}$ .

**Exercise 1.4.** [15] (\*) (Multi-step posterior linearity of  $\xi^\pi$ ) Extend Exercise 1.2 to multi-step histories: show that for finite  $m \geq t$ ,

$$\xi^\pi(\mathfrak{x}_{t:m} \mid \mathfrak{x}_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid \mathfrak{x}_{<t}) \nu^\pi(\mathfrak{x}_{t:m} \mid \mathfrak{x}_{<t}).$$

**Hint:** Apply the general chain rule (Exercise 0.5) to  $\xi(\mathfrak{x}_{1:m}) = \sum_\nu w_\nu \nu(\mathfrak{x}_{1:m})$ , then use factorization (Exercise 0.1).

## 2. Existence of Optimal Policies

Recall that  $V_\nu^*(\mathfrak{x}_{<t}) := \sup_\pi V_\nu^\pi(\mathfrak{x}_{<t})$  is defined as a supremum over all policies. In general, a supremum need not be achieved: for example,  $\sup_{x \in (0,1)} x = 1$ , but no  $x \in (0,1)$  attains this value. This problem shows that in our setting, the supremum *is* attained, so an optimal policy  $\pi_\nu^*$  with  $V_\nu^{\pi_\nu^*} = V_\nu^*$  exists.

**Exercise 2.1.** [05] Consider a single time step. Given history  $\mathfrak{x}_{<t}$  and a function  $Q : \mathcal{A} \rightarrow \mathbb{R}$ , show that  $\sup_\pi \sum_{a \in \mathcal{A}} \pi(a \mid \mathfrak{x}_{<t}) Q(a) = \max_{a \in \mathcal{A}} Q(a)$ , and that the supremum is attained by the deterministic policy that places all probability on an action achieving the maximum.

**Exercise 2.2 (Bellman equation).** [15] Show that for finite  $m$  and  $t \leq m$ :

$$V_\nu^{\pi, m}(\mathfrak{x}_{<t}) = \sum_{\mathfrak{x}'_t} \nu^\pi(\mathfrak{x}'_t \mid \mathfrak{x}_{<t}) \left[ (1 - \gamma) r'_t + \gamma V_\nu^{\pi, m}(\mathfrak{x}_{<t} \mathfrak{x}'_t) \right],$$

**Hint:** Use the general chain rule (Exercise 0.5) to peel off the *first* step,

$$\nu^\pi(\mathfrak{x}_{t:m} \mid \mathfrak{x}_{<t}) = \nu^\pi(\mathfrak{x}_t \mid \mathfrak{x}_{<t}) \cdot \nu^\pi(\mathfrak{x}_{t+1:m} \mid \mathfrak{x}_{1:t}).$$



Break up the sum  $\sum_{\mathfrak{a}_{t:m}} = \sum_{\mathfrak{a}_t} \sum_{\mathfrak{a}_{t+1:m}}$ , and factor out  $r'_t$  using  $\sum_{\mathfrak{a}'_{t+1:m}} \nu^\pi(\mathfrak{a}'_{t+1:m} \mid \mathfrak{a}) = 1$  for any history  $\mathfrak{a}$ .

**Exercise 2.3 (Backward induction).** [20] (\*) Using the Bellman equation from Exercise 2.2 and Exercise 2.1, show by backward induction on  $t = m, m-1, \dots, 1$  that for each finite  $m$ , a deterministic optimal policy exists.

**Exercise 2.4 (Bellman optimality equation).** [10] Using Exercise 2.3, show that for finite  $m$ :

$$V_\nu^{*,m}(\mathfrak{a}_{<t}) = \max_{a'_t} \sum_{e'_t} \nu(e'_t \mid \mathfrak{a}_{<t} a'_t) \left[ (1-\gamma) r'_t + \gamma V_\nu^{*,m}(\mathfrak{a}_{<t} \mathfrak{a}'_t) \right].$$

where  $e'_t = o'_t r'_t$  and  $\mathfrak{a}'_t = a'_t e'_t$ .

**Exercise 2.5.** [15] (\*) (Existence of  $V_\nu^*$ ) Show that the pointwise limit  $V_\nu^*(\mathfrak{a}_{<t}) := \lim_{m \rightarrow \infty} V_\nu^{*,m}(\mathfrak{a}_{<t})$  exists for all histories  $\mathfrak{a}_{<t}$ .

**Hint:** Show  $V_\nu^{*,m+1}(\mathfrak{a}_{<t}) \geq V_\nu^{*,m}(\mathfrak{a}_{<t})$  by expanding  $V_\nu^{*,m+1}$ . Use this and Exercise 1.3 and [Monotone Convergence Theorem](#) to obtain the result.

**Exercise 2.6 (Infinite-horizon Bellman optimality equation).** [10] Show that the Bellman optimality equation (Exercise 2.4) extends to  $m = \infty$ :

$$V_\nu^*(\mathfrak{a}_{<t}) = \max_{a'_t} \sum_{e'_t} \nu(e'_t \mid \mathfrak{a}_{<t} a'_t) \left[ (1-\gamma) r'_t + \gamma V_\nu^*(\mathfrak{a}_{<t} \mathfrak{a}'_t) \right].$$

**Exercise 2.7 (Optimal policy for infinite horizon).** [25] (\*) Show that a deterministic policy  $\pi_\nu^*$  exists achieving  $V_\nu^{\pi_\nu^*}(\mathfrak{a}_{<t}) = V_\nu^*(\mathfrak{a}_{<t})$  for all  $\mathfrak{a}_{<t}$ .

*Approach:*

- Define  $\pi_\nu^*$  as the greedy policy (the arg max action from Exercise 2.6).
- Write two Bellman equations: one for  $V_\nu^*$ , one for  $V_\nu^{\pi_\nu^*,m}$ .
- Define  $\Delta V_\nu^m := V_\nu^* - V_\nu^{\pi_\nu^*,m}$  and obtain a recursive equation for it.
- Iterate out to the finite horizon, noting  $V_\nu^{\pi_\nu^*,m}(\mathfrak{a}_{1:m}) = 0$  (why?)
- Bound the tail, take  $m \rightarrow \infty$ .

### 3. Dominance of the Bayesian Mixture

**Exercise 3.1.** [03] Show that  $\xi(\mathbf{a}_{<t}) \geq w_\nu \cdot \nu(\mathbf{a}_{<t})$  for every  $\nu \in \mathcal{M}$ .

**Exercise 3.2.** [10] Conclude that  $\xi^\pi(\mathbf{a}_{1:m}) \geq w_\nu \cdot \nu^\pi(\mathbf{a}_{1:m})$  for any  $\nu \in \mathcal{M}$  and any policy  $\pi$ .

### 4. Properties of $V_\xi$

**Exercise 4.1.** [15] Show that for finite  $m$ :

$$V_\xi^{\pi,m}(\mathbf{a}_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid \mathbf{a}_{<t}) V_\nu^{\pi,m}(\mathbf{a}_{<t}).$$

**Hint:** Use the multi-step posterior linearity of  $\xi^\pi$  (Exercise 1.4).

**Fact 4.1** (Dominated convergence for sums). *If  $a_{\nu,m} \rightarrow a_\nu$  as  $m \rightarrow \infty$  for each  $\nu$ , and  $|a_{\nu,m}| \leq b_\nu$  for all  $m$  with  $\sum_\nu b_\nu < \infty$ , then  $\sum_\nu a_{\nu,m} \rightarrow \sum_\nu a_\nu$ . (For finite sums this is trivial.)*

**Exercise 4.2 (Infinite-horizon linearity).** [17] (\*) Show that the result extends to  $m = \infty$ :

$$V_\xi^\pi(\mathbf{a}_{<t}) = \sum_{\nu \in \mathcal{M}} w(\nu \mid \mathbf{a}_{<t}) V_\nu^\pi(\mathbf{a}_{<t}).$$

**Hint:** Take  $m \rightarrow \infty$  in the finite-horizon result. You will need Fact 4.1 to exchange limit and sum for countably infinite  $\mathcal{M}$ .

Exercise 4.1 shows  $V_\xi^\pi$  is linear in  $\nu$ . The optimal value  $V_\xi^*$  is only convex: a single policy must perform well across all  $\nu \in \mathcal{M}$  simultaneously, rather than being tailored to each  $\nu$  individually.

**Exercise 4.3.** [10] (\*) (Convexity of  $V_\xi^*$ ) Using Exercise 4.1, show that for finite  $m$ :

$$V_\xi^{*,m}(\mathbf{a}_{<t}) \leq \sum_{\nu \in \mathcal{M}} w(\nu \mid \mathbf{a}_{<t}) V_\nu^{*,m}(\mathbf{a}_{<t}).$$

**Hint:** Apply Exercise 4.1 with  $\pi = \pi_\xi^*$ , the Bayes-optimal policy for  $\xi$ .

**Exercise 4.4.** [15] (\*) (Non-linearity of  $V_\xi^*$ ) Show by example that the inequality in Exercise 4.3 can be strict, i.e.  $V_\xi^{*,m}$  is *not* linear in  $\nu$ .

**Hint:** Consider  $\mathcal{M} = \{\nu_0, \nu_1\}$ : predicting a two-headed coin vs. a two-tailed coin.

## 5. The Expectimax Form of AIXI

We have specified AIXI only implicitly, as the Bayes-optimal policy  $\pi_\xi^*$  (Definition 0.4). We now unroll this definition into the explicit **expectimax** expression: an alternating sequence of maximizations over actions and  $\xi$ -expectations over percepts.

**Exercise 5.1 (Expectimax form of AIXI).** [20] By iterating the finite-horizon Bellman optimality equation (Exercise 2.4), show that

$$V_\xi^{*,m}(\mathfrak{a}_{<t}) = (1-\gamma) \max_{a_t} \sum_{e_t} \xi(e_t | \mathfrak{a}_{<t} a_t) \max_{a_{t+1}} \sum_{e_{t+1}} \xi(e_{t+1} | \mathfrak{a}_{<t+1} a_{t+1}) \cdots \max_{a_m} \sum_{e_m} \xi(e_m | \mathfrak{a}_{<m} a_m) G_{t-1:m}$$

and hence, collecting the percept factors with the chain rule (Exercise 0.2) and taking  $m \rightarrow \infty$  (Exercises 2.5 and 2.7), that AIXI selects the action

$$a_t = \pi_\xi^*(\mathfrak{a}_{<t}) \in \arg \max_{a_t} \lim_{m \rightarrow \infty} \sum_{e_t} \max_{a_{t+1}} \sum_{e_{t+1}} \cdots \max_{a_m} \sum_{e_m} \xi(e_{t:m} | \mathfrak{a}_{<t} a_{t:m}) G_{t-1:m}$$

**Hint:** Prove the first display by backward induction on  $t$  from  $m$  down to 1, using the finite-horizon Bellman optimality equation (Exercise 2.4) and the return recursion  $G_{t-1:m} = r_t + \gamma G_{t:m}$  (Definition 0.3). At each step the leftover reward term  $(1-\gamma)r_t$  is constant in the deeper variables; carry it inward using  $\sum_{e_k} \xi(e_k | \mathfrak{a}_{<k} a_k) = 1$  and  $c + \max(\cdot) = \max(c + \cdot)$ .

## On-Policy Value Convergence

The following problems prove the first two main results: on-policy value convergence (Section 7), and that AIXI can't be fooled in deterministic environments (Section 8). The path to the self-optimizing property (advanced stretch goal) resumes at Section 9.

## 6. Bounding Expectation Differences by Total Variation

**Definition 6.1** (Total variation distance). The **total variation distance** between probability measures  $P$  and  $Q$  on a countable set  $\Omega$  is  $\text{TV}_\Omega(P, Q) := \sup_{S \subseteq \Omega} |P(S) - Q(S)|$ , where  $P(S) := \sum_{\omega \in S} P(\omega)$ .

**Definition 6.2** (Expectation under  $P$ ). For a probability measure  $P$  on a countable set  $\Omega$  and a function  $f : \Omega \rightarrow \mathbb{R}$ , the **expectation** of  $f$  under  $P$  is  $\mathbb{E}_P[f] := \sum_{\omega \in \Omega} f(\omega) P(\omega)$ .

The following technical lemma is needed for the on-policy value convergence proof.

**Exercise 6.1.** [15] (\*) Let  $f : \Omega \rightarrow [0, c]$ . Show that  $|\mathbb{E}_P[f] - \mathbb{E}_Q[f]| \leq c \cdot \text{TV}_\Omega(P, Q)$ .

**Hint:** Define  $A^+ = \{\omega \in \Omega : P(\omega) \geq Q(\omega)\}$  and decompose the expectation difference as a sum over  $A^+$  and its complement  $A^- = \Omega \setminus A^+$ .

## 7. On-Policy Value Convergence of Bayes

The following definitions are needed for the on-policy value convergence theorem.

**Definition 7.1** (Probability of events). A **finite-length event** is a set  $A \subseteq (\mathcal{A} \times \mathcal{E})^t$  of histories of fixed length  $t$ . Its probability under  $\nu^\pi$  is

$$\nu^\pi(A) := \sum_{\mathfrak{a}_{1:t} \in A} \nu^\pi(\mathfrak{a}_{1:t}).$$

An **event on infinite histories** is any set built from finite-length events by countable unions, intersections, and complements.<sup>a</sup> Probabilities of such events are uniquely determined by two properties:

- **Normalization:**  $\nu^\pi((\mathcal{A} \times \mathcal{E})^\infty) = 1$ .
- **Countable additivity:** if  $A_1, A_2, \dots$  are pairwise disjoint events, then  $\nu^\pi(\bigcup_{n=1}^\infty A_n) = \sum_{n=1}^\infty \nu^\pi(A_n)$ .

From these, all standard rules of probability can be derived (e.g.  $\nu^\pi(A \cup B) = \nu^\pi(A) + \nu^\pi(B) - \nu^\pi(A \cap B)$ , etc.), though we will not prove them here. Two derived properties we use explicitly:

- **Complement:**  $\nu^\pi(A^c) = 1 - \nu^\pi(A)$ .
- **Monotone limits:** if  $A_1 \subseteq A_2 \subseteq \dots$ , then  $\nu^\pi(\bigcup_{n=1}^\infty A_n) = \lim_{n \rightarrow \infty} \nu^\pi(A_n)$ .

Further useful notions:

- **Conditional probability:** The probability of event  $A$  given observed history  $\mathfrak{x}_{<t}$  is

$$\nu^\pi(A \mid \mathfrak{x}_{<t}) := \frac{\nu^\pi(\{\mathfrak{x}_{<t}h : h \in A\})}{\nu^\pi(\mathfrak{x}_{<t})}.$$

---

<sup>a</sup>We are deliberately avoiding a formal treatment of measure theory here. Not all subsets of  $(\mathcal{A} \times \mathcal{E})^\infty$  are measurable; we restrict to “nice” (measurable) sets built from finite-prefix conditions via countable set operations, which suffice for everything in this sheet. For a rigorous treatment using  $\sigma$ -algebras and probability measures, see [?, Chapter 2.2].

## Tip

**Example: Example: a fair coin shows 1 infinitely often.** Let  $\mathcal{A} = \mathcal{O} = \{0, 1\}$ ,  $\mathcal{R} = \{0\}$ , and let  $\nu$  be a fair coin that ignores the action:  $\nu(o_t = 1 \mid \mathfrak{x}_{<t} a_t) = \frac{1}{2}$  for all  $\mathfrak{x}_{<t}, a_t$ . Consider the event  $F := “o_t = 1 \text{ for infinitely many } t”$ .

*Step 1: Build  $F^c$  from finite-prefix events.* For each  $t$ , the set  $\{o_t = 0\} := \{\mathfrak{x}_{1:t} : o_t = 0\}$  is a finite-length event. For each  $N$  and  $T \geq N$ , the set  $D_{N,T} := \bigcap_{t=N}^T \{o_t = 0\}$  is also a finite-length event (determined by time  $T$ ). Define  $C_N := \bigcap_{t=N}^\infty \{o_t = 0\}$  (“all zeros from time  $N$  onwards”): this is an infinite-history event, built as a countable intersection of finite-length events. Then  $F^c = \bigcup_{N=1}^\infty C_N$  (“eventually all zeros”).

*Step 2: Compute  $\nu^\pi(C_N) = 0$ .* Since  $D_{N,T} \supseteq D_{N,T+1} \supseteq \dots$  (adding more constraints), the monotone limit property (for decreasing sets) gives:

$$\nu^\pi(C_N) = \lim_{T \rightarrow \infty} \nu^\pi(D_{N,T}) = \lim_{T \rightarrow \infty} \left(\frac{1}{2}\right)^{T-N+1} = 0.$$

(The probability  $\nu^\pi(D_{N,T}) = (1/2)^{T-N+1}$  is independent of  $\pi$ , since  $\nu$  ignores actions.)

*Step 3: By countable additivity:*  $\nu^\pi(F^c) \leq \sum_{N=1}^\infty \nu^\pi(C_N) = 0$ , so  $\nu^\pi(F) = 1$ . A fair coin shows 1 infinitely often, almost surely.

As such,  $F^c$  (the set of all infinite histories containing only finitely many 1s) is a  $\nu^\pi$ -measure-zero set. Such histories “exist” as infinite sequences, but occur with probability zero.

**Definition 7.2** (Covering<sup>1</sup>). We say  $Q$  **covers**  $P$  if  $Q$  is positive everywhere  $P$  is:  $P(A) > 0 \implies Q(A) > 0$  for all events  $A$ . Equivalently: any event that  $Q$  rules out,  $P$  also rules out. By Section 3,  $\xi^\pi$  covers  $\mu^\pi$ .

---

<sup>1</sup>The standard name is *absolute continuity* of  $P$  with respect to  $Q$ , written  $P \ll Q$ .

**Definition 7.3** (Convergence  $\nu^\pi$ -almost surely). Let  $f_t : \mathcal{H}^{t-1} \rightarrow \mathbb{R}$  be a sequence of functions, where  $f_t$  depends on the history  $\mathfrak{x}_{<t}$  of length  $t - 1$ . We write  $f_t(\mathfrak{x}_{<t}) \rightarrow 0$   $\nu^\pi$ -almost surely ( $\nu^\pi$ -a.s.) if

$$\nu^\pi\left(\left\{\mathfrak{x}_{1:\infty} : f_t(\mathfrak{x}_{<t}) \not\rightarrow 0\right\}\right) = 0.$$

This set is built from finite-prefix conditions:  $\{f_t \not\rightarrow 0\} = \bigcup_{n=1}^{\infty} \bigcap_{N=1}^{\infty} \bigcup_{t=N}^{\infty} \{|f_t(\mathfrak{x}_{<t})| > \frac{1}{n}\}$ , so its probability is well-defined (Definition 7.1).

### Tip

**Example: Unpacking “ $f_t \not\rightarrow 0$ ”.** The set  $\{f_t \not\rightarrow 0\}$  looks intimidating, but it reads naturally from the inside out:

- $\{|f_t(\mathfrak{x}_{<t})| > \frac{1}{n}\}$  is a finite-length event: “at time  $t$ , the function is at least  $\frac{1}{n}$  away from zero.”
- $\bigcup_{t=N}^{\infty} \{|f_t| > \frac{1}{n}\}$ : “at *some* time  $t \geq N$ ,  $f_t$  is at least  $\frac{1}{n}$  away from zero.”
- $\bigcap_{N=1}^{\infty} \bigcup_{t=N}^{\infty} \{|f_t| > \frac{1}{n}\}$ : “for *every*  $N$ , there is some  $t \geq N$  where  $f_t$  is at least  $\frac{1}{n}$  from zero”, i.e.,  $f_t$  exceeds  $\frac{1}{n}$  *infinitely often*.
- $\bigcup_{n=1}^{\infty} \bigcap_{N=1}^{\infty} \bigcup_{t=N}^{\infty} \{|f_t| > \frac{1}{n}\}$ : “for *some*  $\frac{1}{n} > 0$ ,  $f_t$  exceeds  $\frac{1}{n}$  infinitely often.”

This last condition is exactly  $\{f_t \not\rightarrow 0\}$ : convergence  $f_t \rightarrow 0$  means that for *every*  $\varepsilon > 0$ ,  $|f_t| \leq \varepsilon$  for all sufficiently large  $t$ . Its negation is that *some*  $\varepsilon > 0$  is exceeded infinitely often.

Each layer is a countable union or intersection of the previous, so the whole set is a well-defined event on infinite histories.

The Bayesian agent uses the mixture  $\xi$  because the true environment  $\mu$  is unknown. A natural question: does planning with  $\xi$  eventually become as good as planning with  $\mu$ ? The following theorem says *yes*: the value of any fixed policy  $\pi$ , as evaluated by  $\xi$ , converges to the value under the true environment  $\mu$ , along histories that  $\mu^\pi$  actually generates.<sup>2</sup> This tells us that the Bayesian mixture “learns” to predict the true environment’s value, on policy.

**Theorem 7.4** (On-policy value convergence; [?, Theorem 7.3.1]). *For any  $\mu \in \mathcal{M}$  and any policy  $\pi$ :  $V_\xi^\pi(\mathfrak{x}_{<t}) - V_\mu^\pi(\mathfrak{x}_{<t}) \rightarrow 0$  as  $t \rightarrow \infty$ ,  $\mu^\pi$ -almost surely. That is, the set of infinite histories along which the value difference does not vanish has  $\mu^\pi$ -probability zero:*

$$\mu^\pi\left(\left\{\mathfrak{x}_{1:\infty} : \lim_{t \rightarrow \infty} (V_\xi^\pi(\mathfrak{x}_{<t}) - V_\mu^\pi(\mathfrak{x}_{<t})) \neq 0\right\}\right) = 0.$$

<sup>2</sup>If the true environment is  $\mu$  then we don’t care about the behaviour of the Bayesian agent on histories that have  $\mu^\pi$ -probability zero: such histories will never be observed anyway.

The proof reduces to a finite-horizon TV bound (Exercise 7.1), a covering argument (Exercise 7.2), and the Blackwell–Dubins theorem (Exercise 7.3). The only ingredient we do not prove is Blackwell–Dubins itself, which we state as a given fact.

**Exercise 7.1 (Finite-horizon TV bound on value difference).** [10] Using Section 6, show that for finite  $m$ :

$$|V_{\xi}^{\pi,m}(\mathfrak{x}_{<t}) - V_{\mu}^{\pi,m}(\mathfrak{x}_{<t})| \leq \text{TV}_{\mathcal{H}^{m-t+1}}(\xi^{\pi}(\cdot | \mathfrak{x}_{<t}), \mu^{\pi}(\cdot | \mathfrak{x}_{<t})),$$

where  $\mathcal{H}^{m-t+1} := (\mathcal{A} \times \mathcal{E})^{m-t+1}$  is the set of all future history segments  $\mathfrak{x}_{t:m}$ .

**Exercise 7.2 (Covering).** [05] Show that  $\xi^{\pi}$  covers  $\mu^{\pi}$  (Definition 7.2).

**Hint:** Use Section 3.

To complete the proof, we need the TV distance on finite segments to vanish  $\mu^{\pi}$ -a.s. as  $t \rightarrow \infty$ . This follows from the following deep result, which we state without proof:

**Fact 7.5** (Blackwell–Dubins merging of opinions [?]). *If  $Q$  covers  $P$ , then  $\sup_S |P(S | \mathfrak{x}_{<t}) - Q(S | \mathfrak{x}_{<t})| \rightarrow 0$   $P$ -almost surely, where the supremum ranges over all measurable events  $S$  (including events on infinite histories).*

The proof of Blackwell–Dubins requires the Radon–Nikodym derivative and Lévy’s martingale convergence theorem: tools from measure theory that are beyond the scope of this sheet. See [?, Chapter 3.9] for discussion.

**Exercise 7.3 (On-policy convergence).** [15] Using Fact 7.5, Exercise 7.1 and Exercise 7.2, prove Theorem 7.4.

**Exercise 7.4.** [40] (\*) Prove Fact 7.5.

**Hint:** See [?]. The proof uses the Radon–Nikodym derivative  $dP/dQ$ , Lévy’s martingale convergence theorem, and the Lebesgue decomposition. No elementary proof is known that avoids graduate-level measure theory.

## 8. AIXI Cannot Be Fooled in Deterministic Environments

**Theorem 8.1** (AIXI cannot act poorly in good environments). *If  $\mu \in \mathcal{M}$  is deterministic and  $V_{\mu}^*(\mathfrak{x}_{<t}) > \varepsilon > 0$  for all  $t$  along the history generated by  $\mu$  and  $\pi_{\xi}^*$ , then  $V_{\xi}^*(\mathfrak{x}_{<t}) \geq w_{\mu} \varepsilon > 0$  for all  $t$ . [?, Theorem 7.4.10]*

**Exercise 8.1.** [10] Show that  $V_{\xi}^*(\mathfrak{a}_{<t}) \geq w(\mu \mid \mathfrak{a}_{<t}) \cdot V_{\mu}^*(\mathfrak{a}_{<t})$ .

**Hint:** Use linearity of  $V^{\pi}$  (Exercise 4.2).

**Exercise 8.2.** [15] Show  $w(\mu \mid \mathfrak{a}_{<t}) \geq w_{\mu}$  when  $\mu$  is deterministic, and combine with Exercise 8.1 to prove Theorem 8.1.

*Remark.* With the Solomonoff prior,  $w_{\mu} = 2^{-K(\mu)}$  may be astronomically small. The result guarantees non-zero value, not near-optimal value.

## Sections 9–11: Self-Optimizing Policy (Finite Model Class)

We now prove: if *any* policy can learn to act optimally,  $\pi_{\xi}^*$  also learns. We restrict to finite  $\mathcal{M} = \{\nu_1, \dots, \nu_N\}$  where each  $\nu_i$  is a proper probability measure.

### 9. Likelihood Ratios Are Martingales

The following definition and theorem state the main goal of Sections 9–11.

**Definition 9.1** (Self-optimizing). Fix a historic policy  $\pi$ . A policy  $\tilde{\pi}$  is **self-optimizing** for  $\mathcal{M}$  with respect to  $\pi$  if for every  $\nu \in \mathcal{M}$ :  $V_{\nu}^*(\mathfrak{a}_{<t}) - V_{\nu}^{\tilde{\pi}}(\mathfrak{a}_{<t}) \rightarrow 0$  as  $t \rightarrow \infty$ ,  $\nu^{\pi}$ -almost surely.

The percepts  $e_{1:\infty}$  are sampled from  $\nu$ ; the historic actions  $a_{<t}$  come from  $\pi$ , which may differ from  $\tilde{\pi}$ .

**Theorem 9.2** (Self-optimizing; [?, Theorem 7.5.2]). *Let  $\mathcal{M}$  be finite and fix a historic policy  $\pi$ . If there exists a  $\tilde{\pi}$  self-optimizing for  $\mathcal{M}$  with respect to  $\pi$ , then  $\pi_{\xi}^*$  is also self-optimizing for  $\mathcal{M}$  with respect to  $\pi$ : for any  $\mu \in \mathcal{M}$ ,  $V_{\mu}^*(\mathfrak{a}_{<t}) - V_{\mu}^{\pi_{\xi}^*}(\mathfrak{a}_{<t}) \rightarrow 0$   $\mu^{\pi}$ -almost surely.*

**Definition 9.3** (Supermartingale). A sequence of functions  $X_t : \mathcal{H}^t \rightarrow \mathbb{R}_{\geq 0}$  (each depending on the history  $\mathfrak{a}_{1:t}$ ) is a  $\mu^{\pi}$ -**supermartingale** if  $\mathbb{E}_{\mu^{\pi}}[X_t \mid \mathfrak{a}_{<t}] \leq X_{t-1}$  for all  $t$ ; that is,

$$\sum_{\mathfrak{a}_t} \pi(a_t \mid \mathfrak{a}_{<t}) \mu(e_t \mid \mathfrak{a}_{<t} a_t) X_t(\mathfrak{a}_{1:t}) \leq X_{t-1}(\mathfrak{a}_{<t}).$$

If equality holds, it is a  $\mu^{\pi}$ -**martingale**.

**Intuition.** A supermartingale is a quantity whose expected future value, given the present, is no larger than its current value: it cannot “drift upward on average.” A martingale is the equality case: expected future value equals current value. The key



example is the **likelihood ratio**  $X_{\nu,t}(\mathfrak{a}_{1:t}) := \nu(\mathfrak{a}_{1:t})/\mu(\mathfrak{a}_{1:t})$ : how much more (or less) likely the observed history is under  $\nu$  than under the true environment  $\mu$ . Under  $\mu$ , this ratio is a martingale: the true environment does not, on average, favour any alternative  $\nu$  over itself. Non-negativity plus the (super)martingale structure forces  $X_{\nu,t}$  to converge (by Doob's theorem below), which is the engine behind the self-optimizing proof: it lets us separate environments that remain plausible ( $X_{\nu,\infty} > 0$ ) from those that are eventually ruled out ( $X_{\nu,\infty} = 0$ ), and handle each case differently in Section 11.

**Fact 9.4** (Doob's supermartingale convergence). *If  $(X_t)_{t \geq 0}$  is a non-negative  $\mu^\pi$ -supermartingale, then there exists a function  $X_\infty : \mathcal{H}^\infty \rightarrow \mathbb{R}_{\geq 0}$  of the infinite history such that  $X_t(\mathfrak{a}_{1:t}) \rightarrow X_\infty(\mathfrak{a}_{1:\infty})$  as  $t \rightarrow \infty$ , with  $X_\infty(\mathfrak{a}_{1:\infty}) < \infty$ , for  $\mu^\pi$ -almost every infinite history  $\mathfrak{a}_{1:\infty}$ . That is:*

$$\mu^\pi\left(\left\{\mathfrak{a}_{1:\infty} \in (\mathcal{A} \times \mathcal{E})^\infty : X_t(\mathfrak{a}_{1:t}) \not\rightarrow X_\infty(\mathfrak{a}_{1:\infty})\right\}\right) = 0.$$

For each  $\nu \in \mathcal{M}$ , define the likelihood ratio

$$X_{\nu,t} : \mathcal{H}^t \rightarrow \mathbb{R}_{\geq 0}, \quad X_{\nu,t}(\mathfrak{a}_{1:t}) := \frac{\nu(\mathfrak{a}_{1:t})}{\mu(\mathfrak{a}_{1:t})}, \quad X_{\nu,0} := 1.$$

$X_{\nu,t}$  is a *function* of the history  $\mathfrak{a}_{1:t}$ , not a number: different histories give different values. Throughout this section and the next, any unqualified statement of the form “ $X_{\nu,t} \geq c$ ” or “ $X_{\nu,t} \rightarrow X_{\nu,\infty}$ ” is shorthand for “ $X_{\nu,t}(\mathfrak{a}_{1:t}) \geq c$ ” or “ $X_{\nu,t}(\mathfrak{a}_{1:t}) \rightarrow X_{\nu,\infty}(\mathfrak{a}_{1:\infty})$ ” holding  $\mu^\pi$ -almost surely, i.e. on every infinite history outside a set of  $\mu^\pi$ -measure zero. We will not track these null sets explicitly; their countable union over all claims is still a  $\mu^\pi$ -null set.

**Exercise 9.1.** [15] Show that  $X_{\nu,t}$  is a  $\mu^\pi$ -martingale:  $\mathbb{E}_{\mu^\pi}[X_{\nu,t} \mid \mathfrak{a}_{<t}] = X_{\nu,t-1}$ .

**Hint:** Write  $X_{\nu,t}(\mathfrak{a}_{1:t}) = X_{\nu,t-1}(\mathfrak{a}_{<t}) \cdot \nu(e_t \mid \mathfrak{a}_{<t}a_t)/\mu(e_t \mid \mathfrak{a}_{<t}a_t)$ . Sum over  $a_t, e_t$ ;  $\mu$  cancels, leaving  $\sum_{e_t} \nu(e_t \mid \mathfrak{a}_{<t}a_t) = 1$ .

**Exercise 9.2.** [05] Apply Fact 9.4 to conclude  $X_{\nu,t} \rightarrow X_{\nu,\infty} < \infty$   $\mu^\pi$ -a.s.

**Exercise 9.3.** [10] Define  $X_{\xi,t} := \xi(\mathfrak{a}_{1:t})/\mu(\mathfrak{a}_{1:t})$ . Show  $X_{\xi,t} = \sum_\nu w_\nu X_{\nu,t}$  and  $X_{\xi,t} \geq w_\mu > 0$ .

## 10. Change of Measure

**Exercise 10.1.** [15] Let  $A$  be a set of finite histories of length  $m$ . Show:

$$\nu^\pi[\mathfrak{a}_{1:m} \in A] = \mathbb{E}_{\mu^\pi}[X_{\nu,m} \cdot \mathbb{I}[\mathfrak{a}_{1:m} \in A]].$$

**Hint:** Use Exercise 0.1 to show  $\nu^\pi/\mu^\pi = \nu/\mu = X_{\nu,m}$ .

**Exercise 10.2.** (Given.) The identity extends to infinite histories: for any event  $A \subseteq (\mathcal{A} \times \mathcal{E})^\infty$ ,

$$\nu^\pi[A] = \mathbb{E}_{\mu^\pi}[X_{\nu,\infty} \cdot \mathbb{I}_{\{\mathfrak{a}_{1:\infty} \in A\}}].$$

This says that  $X_{\nu,\infty}$  plays the role of the Radon–Nikodym derivative  $d\nu^\pi/d\mu^\pi$  on the space of infinite histories. The proof rests on two ingredients beyond the scope of this worksheet:

- **$L^1$  martingale convergence** (closed martingale / Lévy’s upward theorem): since  $\mathbb{E}_{\mu^\pi}[X_{\nu,t}] = 1$  for every  $t$ , the non-negative  $\mu^\pi$ -martingale  $(X_{\nu,t})$  is uniformly integrable, so  $X_{\nu,t} \rightarrow X_{\nu,\infty}$  in  $L^1(\mu^\pi)$ , not merely  $\mu^\pi$ -a.s. This lets us pass the  $t \rightarrow \infty$  limit through the expectation.
- **Uniqueness of measure extension** (Carathéodory /  $\pi$ - $\lambda$  theorem): both sides of the identity are finite measures on  $(\mathcal{A} \times \mathcal{E})^\infty$ ; the finite-history identity (Exercise 10.1) shows they agree on all cylinder events  $A = A_0 \times (\mathcal{A} \times \mathcal{E})^\infty$ , and cylinders generate the full event  $\sigma$ -algebra, so agreement extends uniquely to every event.

We omit the proof and use this identity freely below.

**Exercise 10.3 (Markov inequality for change of measure).** [15] Let  $E$  be an event of infinite histories. For any  $\varepsilon > 0$ , show that

$$\mu^\pi[E \cap \{X_{\nu,\infty} \geq \varepsilon\}] \leq \frac{\nu^\pi[E]}{\varepsilon}.$$

**Hint:** On  $E \cap \{X_{\nu,\infty} \geq \varepsilon\}$ ,  $X_{\nu,\infty} \geq \varepsilon$  pointwise. Take  $\mu^\pi$ -expectations and apply Exercise 10.2.

## 11. Proving the Self-Optimizing Property

Fix a policy  $\tilde{\pi}$  that is self-optimizing for  $\mathcal{M}$  with respect to  $\pi$  (Definition 9.1): its existence is the hypothesis of Theorem 9.2. For each  $\nu \in \mathcal{M}$ , define the **suboptimality gap**

$$\delta_{\nu,t} : \mathcal{H}^{t-1} \rightarrow [0, 1], \quad \delta_{\nu,t}(\mathfrak{a}_{<t}) := V_\nu^*(\mathfrak{a}_{<t}) - V_{\nu}^{\tilde{\pi}}(\mathfrak{a}_{<t}).$$

As with  $X_{\nu,t}$ , we write  $\delta_{\nu,t}$  without its argument when convenient: “ $\delta_{\nu,t} \rightarrow 0$   $\mu^\pi$ -a.s.” means  $\delta_{\nu,t}(\mathfrak{a}_{<t}) \rightarrow 0$  on every infinite history outside a  $\mu^\pi$ -null set, and similarly for “ $\delta_{\nu,t} \leq 1$ ” etc.

**Exercise 11.1 (Chain of inequalities).** [20] Show:

$$0 \leq w(\mu \mid \mathfrak{a}_{<t})[V_\mu^* - V_\mu^{\pi_\xi^*}] \leq \sum_{\nu \in \mathcal{M}} w(\nu \mid \mathfrak{a}_{<t}) \delta_{\nu,t}.$$

**Hint:** Use  $V_\xi^{\pi_\xi^*} \geq V_\xi^{\tilde{\pi}}$  and Exercise 4.2.

**Exercise 11.2 (Vanishing posterior).** [10] Show: if  $X_{\nu,\infty} = 0$  then  $w(\nu \mid \mathfrak{a}_{<t}) \rightarrow 0$   $\mu^\pi$ -a.s.

**Hint:** Show  $w(\nu \mid \mathfrak{a}_{<t}) = w_\nu X_{\nu,t-1} / X_{\xi,t-1}$  and use Exercise 9.3.

**Exercise 11.3 (Transferring convergence).** [20] Let  $F := \{\mathfrak{a}_{1:\infty} : \delta_{\nu,t}(\mathfrak{a}_{<t}) \not\rightarrow 0\}$ , the set of infinite histories on which the suboptimality gap fails to vanish. Show that  $\mu^\pi[F \cap \{X_{\nu,\infty} > 0\}] = 0$ .

**Hint:** Self-optimizing (Definition 9.1) gives  $\nu^\pi[F] = 0$ . Apply Exercise 10.3 with  $E = F$  and  $\varepsilon = 1/n$  to conclude  $\mu^\pi[F \cap \{X_{\nu,\infty} \geq 1/n\}] = 0$  for each  $n \geq 1$ . Then note  $\{X_{\nu,\infty} > 0\} = \bigcup_{n \geq 1} \{X_{\nu,\infty} \geq 1/n\}$  and take the countable union (Definition 7.1).

**Exercise 11.4 (Single-term convergence).** [10] Combine Exercise 11.2 and Exercise 11.3 to show  $w(\nu \mid \mathfrak{a}_{<t}) \delta_{\nu,t} \rightarrow 0$   $\mu^\pi$ -a.s.

**Exercise 11.5 (Self-Optimizing Theorem).** [15] Combine Exercise 11.4, Exercise 11.1, Exercise 9.3 to finally prove the main result of Theorem 9.2.

**Hint:**  $w(\mu \mid \mathfrak{a}_{<t}) = w_\mu / X_{\xi,t-1}$ .

*Remark.* We do not need to know which policy  $\tilde{\pi}$  is self-optimizing, or whether it is computable. Mere existence suffices. For countable  $\mathcal{M}$ , this final step is harder: a countable sum of  $\mu^\pi$ -a.s.-convergent sequences need not converge  $\mu^\pi$ -a.s. The general proof uses a “convergence of mixture tails” argument [?, Lem. 5.28].

## A. Worked Example: Bayesian Mixture

### Note

#### Setup.

- **Actions:**  $\mathcal{A} = \{H, T\}$  (predict the next coin flip)
- **Observations:**  $\mathcal{O} = \{H, T\}$  (actual coin flip)
- **Rewards:**  $\mathcal{R} = \{0, 1\}$ , with  $r_t = \llbracket a_t = o_t \rrbracket$
- **Model class:**  $\mathcal{M} = \{\nu_{HH}, \nu_{HT}\}$  (two-headed coin, fair coin)
- **Prior:**  $w_{\nu_{HH}} = w_{\nu_{HT}} = \frac{1}{2}$

Before any interaction ( $t = 1$ ,  $\mathfrak{x}_{<1} = \epsilon$ ):

$$\xi(o_1 = H \mid a_1) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}.$$

After observing a head ( $t = 2$ ), the posterior updates:

$$w(\nu_{HH} \mid \mathfrak{x}_1) = \frac{1}{2} \cdot \frac{1}{3/4} = \frac{2}{3}, \quad w(\nu_{HT} \mid \mathfrak{x}_1) = \frac{1}{2} \cdot \frac{1/2}{3/4} = \frac{1}{3}.$$

Updated prediction:  $\xi(o_2 = H \mid \mathfrak{x}_1 a_2) = \frac{2}{3} \cdot 1 + \frac{1}{3} \cdot \frac{1}{2} = \frac{5}{6}$ .

## B. Worked Example: Value Function

Continuing from [Appendix A](#). Suppose the agent always predicts  $H$  (policy  $\pi_H$ ).

Under  $\nu_{HH}$ : always correct,  $r_t = 1$  every step:  $V_{\nu_{HH}}^{\pi_H}(\epsilon) = (1 - \gamma) \sum_{k=0}^{\infty} \gamma^k \cdot 1 = 1$ .

Under  $\nu_{HT}$ : correct half the time:  $V_{\nu_{HT}}^{\pi_H}(\epsilon) = (1 - \gamma) \sum_{k=0}^{\infty} \gamma^k \cdot \frac{1}{2} = \frac{1}{2}$ .

The mixture value (by [Exercise 4.2](#)) is:  $V_{\xi}^{\pi_H}(\epsilon) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}$ .

As the agent observes more heads ( $\mu = \nu_{HH}$ ), the posterior on  $\nu_{HH}$  increases towards 1, and  $V_{\xi}^{\pi_H} \rightarrow V_{\nu_{HH}}^{\pi_H} = 1$ . This is on-policy value convergence ([Theorem 7.4](#)) in action.

## C. Knuth's Difficulty Scale

Each subproblem carries a difficulty rating in square brackets, following Knuth's rating scheme for exercises [?] in slightly adapted form. The rating assumes that the material in the preceding problems (on which the subproblem depends) has been understood. In-between values are possible.

[00] *Very easy*. Solvable from the top of your head.

[10] *Easy*. Needs 15 minutes to think, possibly pencil and paper.

[20] *Average*. May take 1–2 hours to answer completely.

[30] *Moderately difficult or lengthy*. May take several hours to a day.

[40] *Quite difficult or lengthy*. Often a significant research result.

[50] *Open research problem*. An obtained solution should be published.

Problems marked (\*) are enrichment: they are off the critical path to the three main results (Theorem 7.4, Theorem 8.1, Theorem 9.2) and can be skipped on a first pass without loss of continuity.